

Why global SBC vendors are watching India closely

- Growing demand for edge AI over cloud dependence
- Cost-sensitive but performance-driven market
- Increasing preference for standard platforms with customisation
- Long deployment lifecycles (10-15 years in many projects)

For vendors like ADLINK, Advantech, and Kontron, India represents not just a sales market but a deployment laboratory for rugged, scalable edge systems.

designs that combine CPUs for system orchestration, GPUs for graphics and compute-intensive vision workloads, and NPUs or dedicated AI engines for low-power, always-on inference. What was once an optional enhancement has now become a baseline expectation for modern SBC platforms.

Platforms such as ADLINK's SBC-35 MTL, based on Intel Core Ultra processors, exemplify this shift by integrating CPU, GPU, and NPU resources directly on a single board. This enables on-board AI inference without relying on discrete accelerators, resulting in lower latency, reduced power consumption, and simpler system integration. As Venkatesh explains, "ADLINK follows a hybrid compute concept in which CPU, GPU, and NPU resources are allocated dynamically based on application needs, whether for video analytics, cybersecurity, sensor fusion, or predictive industrial systems. Entry-level inference workloads can run efficiently on NPUs, while GPUs handle high-end, real-time video analytics".

A similar architectural philosophy is now visible across both x86 and ARM-based SBCs. Intel Core Ultra-based platforms, such as the AAEON PICO-MTU4, extend this hybrid compute model into ultra-compact form factors, delivering meaningful AI performance in space-constrained designs without external accelerators.

On the ARM side, offerings such as Advantech's RSB-3810 using MediaTek's Genio processors, PHYTEC's phyBOARD-Pollux based on NXP's i.MX 8M Plus processors, and SECO's ARM-based Qualcomm

solutions, including its SMARC-based SBC offerings, provide NPU cores that enable efficient inference at significantly lower power levels.

At the high end of the scale, the most popular choice for developing robots and autonomous devices requiring exceedingly high TOPS, often due to multiple high-resolution camera streams, is the Nvidia Jetson AGX Orin platform. This is because its GPU-based artificial intelligence capabilities are better suited to applications that exceed the reach of NPUs integrated into SoCs.

Since edge platforms continue to shrink while workloads escalate, the 3.5-inch (8.8cm) SBC form factor has proven to be an important compromise for product designers. It remains compact, fitting constrained products, while being strong in inference, machine vision, and I/O-rich edge use cases.

Boards such as ADLINK's SBC-35 MTL and SBC-35 ASL illustrate a broader industry push to balance rising compute demands with strict thermal limits, fanless operation, and rich connectivity. These platforms are designed to incorporate the latest processor technology, including AI acceleration, enabling real-time analytics and image recognition workloads without requiring migration to larger, cooling-enhanced configurations.

This momentum is not limited to one vendor. The emphasis the Kontron SBC-AML/ADN 3.5-inch (8.8cm) SBC places on reliability and availability for industrial applications, including transportation, infrastructure, and factory automation, is noteworthy. The focus of DFI's GHF51 series on

space-constrained automation and medical platforms, where x86 longevity and reliability support changing workloads without hardware redesign, is likewise noteworthy.

Despite their small dimensions, even the newest 3.5-inch-class (8.8cm) SBCs can offer surprisingly rich I/O functionality. USB ports, multi-port Ethernet, PCIe support, Wi-Fi and/or LTE, digital I/O, and multi-camera interfaces can all be included, offering substantial opportunities to integrate multiple sensors and displays into a single embedded system.

This is important when developing edge AI systems that need to integrate data from multiple cameras, sensors, and/or networks.

What matters most is not the size of the board from a mechanical perspective, but how much functionality can be packed into that footprint. It has become possible to utilise more capable and robust functionality without changing mechanical designs or thermal architectures. This continuity reduces system design risks, lowers costs, and minimises schedule risks associated with migrating to a new enclosure, cooling solution, or platform, positioning the 3.5-inch (8.8cm) SBC as the default platform for next-generation edge deployments.

Modularity is replacing custom hardware redesign

Another quiet but critical transformation reshaping the SBC landscape is the shift towards modular expansion. Instead of redesigning an entire board for every new project, SBC vendors are increasingly building platform ecosystems around function modules, mezzanine cards, M.2 AI accelerators, and PCIe-based expansion. This approach allows systems to scale and adapt without disturbing the core compute platform.

ADLINK's Function Module (FM) strategy is a clear example of